18B
CHEMICAL THERMODYNAMICS

UNIT 03: WHAT REACTIONS GIVE

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CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 18

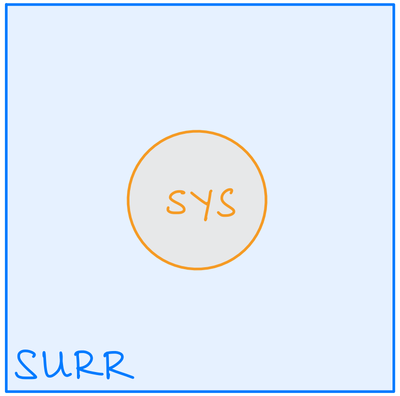
1. State the second law of thermodynamics.
2. Define entropy (S) and identify the sign of ΔS from chemical equations or changes of state.
3. State the third law of thermodynamics.
4. Calculate ΔS_{rxn} from molar entropies.
5. Compare and contrast ΔS_{univ}, ΔS_{sys}, and ΔS_{surr}.
6. Define Gibbs free energy (G) and describe how the signs of ΔH and ΔS affect the sign of ΔG.
7. Relate the sign of ΔG to the favorability/spontaneity of a reaction.
8. Determine the equilibrium temperature of a reaction.
9. Apply the relationship between $\Delta G°$ and the equilibrium constant (K).

Entropy and Temperature

18.3

Second Law of Thermodynamics

- A *spontaneous process* is one that results in an overall *increase* in the entropy (disorder) of the universe. $\Delta S_{univ} > 0$
- Since $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr}$, this means that



		ΔS_{univ}	ΔS_{sys}	ΔS_{surr}	
Spontaneous	✓	+	+	+	
Spontaneous	✓	+	+	−	if $\lvert\Delta S_{sys}\rvert > \lvert\Delta S_{surr}\rvert$
Spontaneous	✓	+	−	+	if $\lvert\Delta S_{sys}\rvert < \lvert\Delta S_{surr}\rvert$
Nonspontaneous	✗	−	−	−	

Gibbs Free Energy

18.4

Unfortunately, the Second Law is practically useless because it is often immeasurable.

$\Delta S_{univ} > 0$

So, how can the spontaneity of a process/reaction be predicted?

- *Exothermic* processes ($\Delta H < 0$) are *more likely* to be spontaneous than endothermic ($\Delta H > 0$) processes.
- Processes where there is an *increase* in entropy ($\Delta S > 0$) are *more likely* to be spontaneous than processes with decreases in entropy ($\Delta S < 0$).

✓ favorable $\Delta H_{sys} < 0$, $\Delta S_{sys} > 0$

✗ unfavorable $\Delta H_{sys} > 0$, $\Delta S_{sys} < 0$

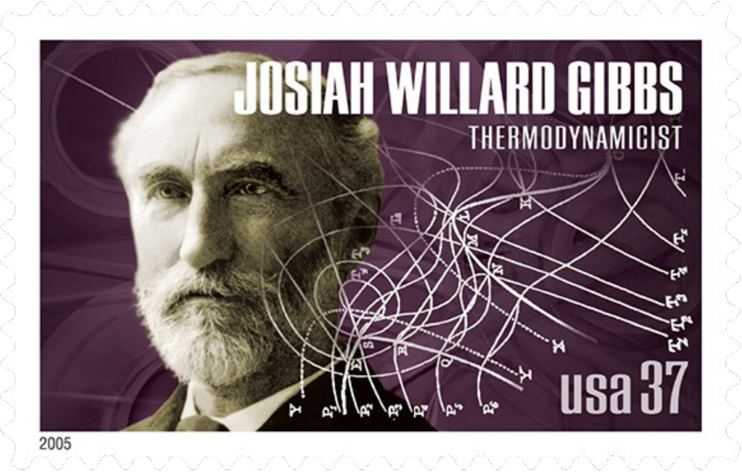
Gibbs Free Energy

18.4

Gibbs Free Energy (G)

- A mathematical model to predict reaction spontaneity in terms of ΔH_{rxn}, ΔS_{rxn}, and T (temperature).
- This new state function is defined as

$$\Delta G_{rxn} = \Delta H_{rxn} - T\Delta S_{rxn}$$



- The sign of ΔG_{rxn} is *opposite* the sign of ΔS_{univ}.
- Most importantly, under constant pressure conditions (i.e., benchtop reactions):

✓ spontaneous favorable $\longrightarrow \Delta S_{univ} > 0$ (+) $\longrightarrow \Delta G_{rxn} < 0$ (−)

✗ nonspontaneous unfavorable $\longrightarrow \Delta S_{univ} < 0$ (−) $\longrightarrow \Delta G_{rxn} > 0$ (+)

Recall the term "driving force" from earlier

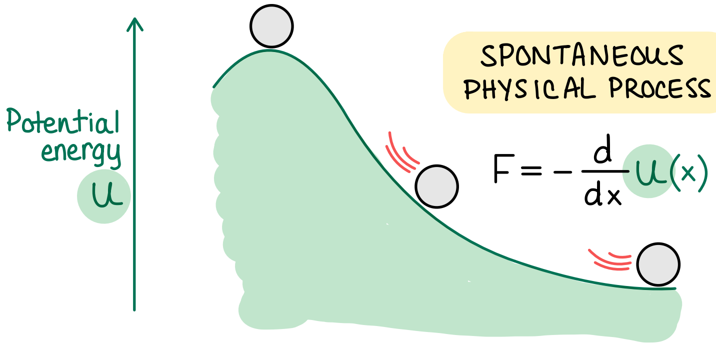
Chemistry in Context

In Chapter 4, we said that reactions occur *spontaneously* due to a combination of changes in heat energy (ΔH_{rxn}) and randomness (ΔS_{rxn}).

We also noted that certain *patterns* of reactants and products tend to be associated with *spontaneous reactions* and classified these patterns as *driving forces*.

1. Formation of a precipitate → Double-replacement reactions
2. Formation of water or neutralization → Acid-base reaction
3. Oxidation-reduction → Combustion, synthesis, decomposition, and single-replacements reactions

But why the term **driving force**? To mimic mechanical/physical systems!



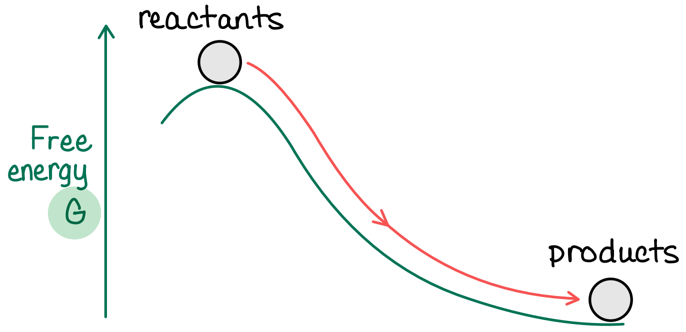
SPONTANEOUS PHYSICAL PROCESS

$$F = -\frac{d}{dx}U(x)$$

spontaneous when force (energy derivative) negative and final state is energy-minimized

SPONTANEOUS CHEMICAL PROCESS

$$F = -\frac{d}{dx}G(x)$$



Gibbs Free Energy

18.4

$$\underbrace{aA + bB}_{\text{reactants}} \rightarrow \underbrace{cC + dD}_{\text{products}}$$

n_R n_P

Calculating changes in Gibbs free energy (ΔG^{o}_{rxn}) from tabulated ΔG^{o}_{f} values.

$$① \quad \Delta G^{o}_{rxn} = \underbrace{\sum n_P \cdot \Delta G^{o}_{f,P}}_{\text{products}} - \underbrace{\sum n_R \cdot \Delta G^{o}_{f,R}}_{\text{reactants}}$$

– or –

$$② \quad \Delta G^{o}_{rxn} = \Delta H^{o}_{rxn} - T\Delta S^{o}_{rxn}$$

Calculating changes in enthalpy (ΔH^{o}_{rxn}) from tabulated ΔH^{o}_{f} values.

$$\Delta H^{o}_{rxn} = \underbrace{\sum n_P \cdot \Delta H^{o}_{f,P}}_{\text{products}} - \underbrace{\sum n_R \cdot \Delta H^{o}_{f,R}}_{\text{reactants}} = \left(c \cdot \Delta H^{o}_{f,C} + d \cdot \Delta H^{o}_{f,D}\right) - \left(a \cdot \Delta H^{o}_{f,A} + b \cdot \Delta H^{o}_{f,B}\right)$$

Calculating changes in entropy (ΔS^{o}_{rxn}) from tabulated S^{o} values.

$$\Delta S^{o}_{rxn} = \underbrace{\sum n_P \cdot S^{o}_{P}}_{\text{products}} - \underbrace{\sum n_R \cdot S^{o}_{R}}_{\text{reactants}} = \left(c \cdot S^{o}_{C} + d \cdot S^{o}_{D}\right) - \left(a \cdot S^{o}_{A} + b \cdot S^{o}_{B}\right)$$

Free-Energy Changes and Temperature

18.5

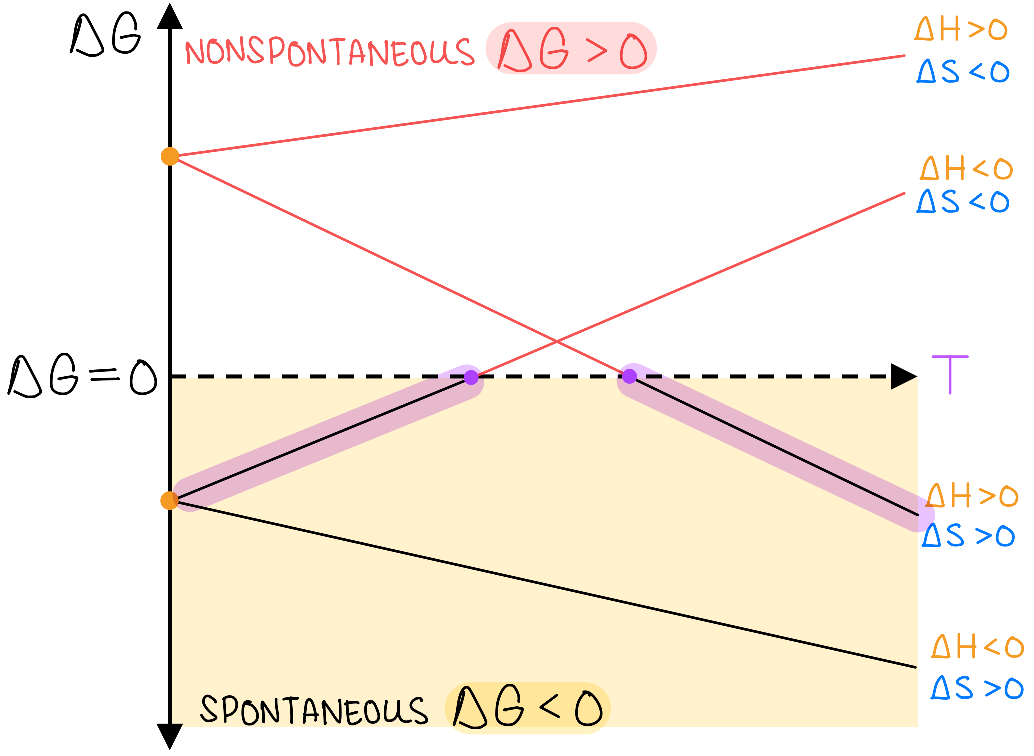
$$\Delta G_{rxn} = \Delta H_{rxn} - T\Delta S_{rxn}$$

$$\underbrace{\Delta G_{rxn}}_{y} = \underbrace{(-\Delta S_{rxn})}_{m}\underbrace{T}_{x} + \underbrace{\Delta H_{rxn}}_{b}$$

For some processes, the value of ΔG_{rxn} is *temperature-dependent*.

So, these processes exhibit *temperature-dependent spontaneity*.

ΔH_{rxn}	ΔS_{rxn}	ΔG_{rxn}	Reaction is ...
–	+	–	spontaneous at all T
+	–	+	nonspontaneous at all T
–	–	–	spontaneous at LOW T
		+	nonspontaneous at HIGH T
+	+	–	spontaneous at HIGH T
		+	nonspontaneous at LOW T



Free-Energy Changes and Temperature

18.5

At equilibrium: $\text{Rate}_{\text{forward}} = \text{Rate}_{\text{reverse}}$

$$Q = K$$

$$\Delta G_{rxn} = 0$$

The "low" and "high" temperatures are defined by solving for T.

$$\Delta G_{rxn} = \Delta H_{rxn} - T\Delta S_{rxn}$$

$$0 = \Delta H_{rxn} - T\Delta S_{rxn}$$

$$\boxed{T = \frac{\Delta H_{rxn}}{\Delta S_{rxn}}}$$

This is useful for calculating the temperature at which phase transitions occur!

IN-CLASS QUESTION 1

Learning Objective(s): 18.8

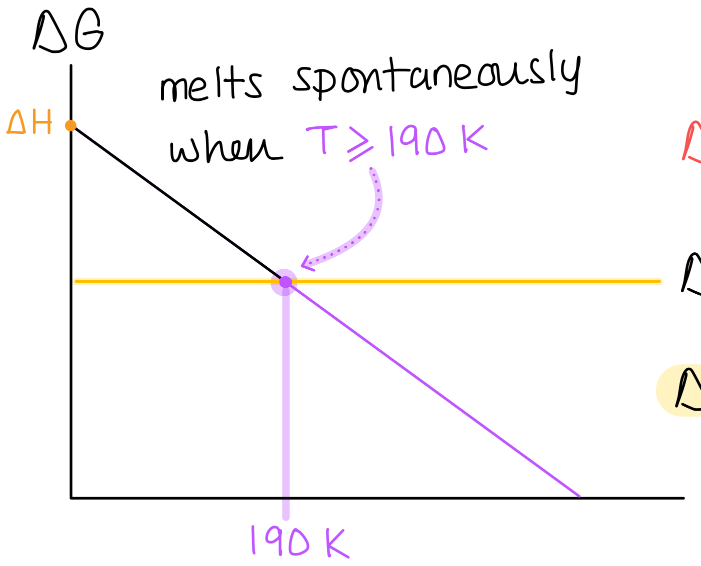
What is the melting point (in K) of an unknown compound A given the standard changes in enthalpy and entropy for the process.

pay attention to units!

$$A(s) \rightarrow A(\ell) \qquad \Delta H° = +8.1\ \frac{kJ}{mol} \qquad \Delta S° = +42.4\ \frac{J}{mol \cdot K}$$

$+$ $\quad$ $+$

$$\underbrace{\Delta G_{rxn}}_{y} = \underbrace{(-\Delta S_{rxn})}_{m}\underbrace{T}_{x} + \underbrace{\Delta H_{rxn}}_{b}$$



$\Delta G > 0$ nonspont

$\Delta G = 0$

$\Delta G < 0$ spont

$$T = \frac{\Delta H°_{rxn}}{\Delta S°_{rxn}}$$

$$= \frac{8.1 \times 10^3\ \frac{J}{mol}}{42.4\ \frac{J}{mol \cdot K}}$$

$$T = 190\ K$$

IN-CLASS QUESTION 2

Learning Objective(s): 18.6, 18.7

$\Delta H_{rxn} > 0$

At a given temperature and pressure, the following reaction is endothermic.

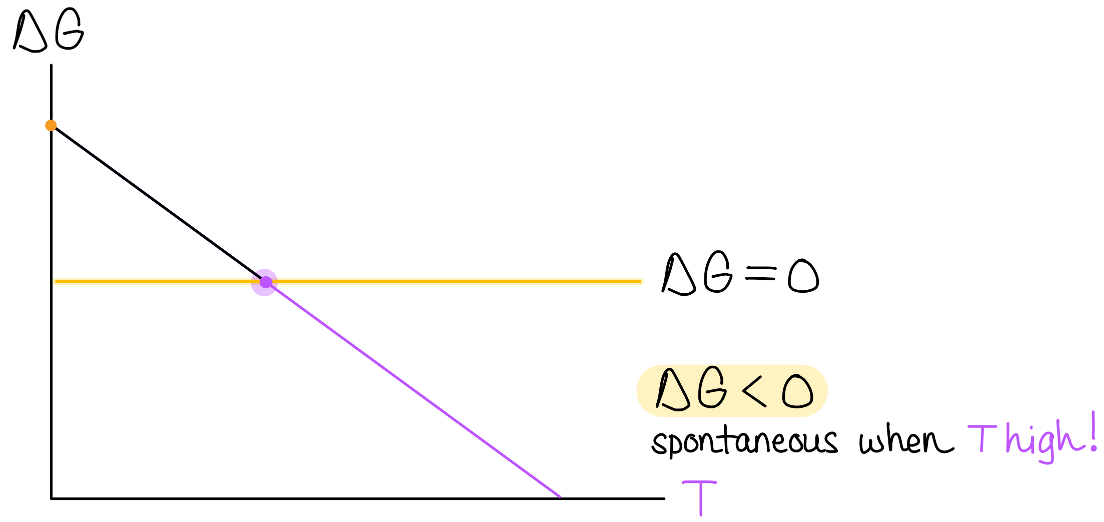
$$N_2O_4(g) \rightleftharpoons 2\ NO_2(g)$$

$\Delta S_{rxn} > 0$ because $\Delta n_{gas} > 0$

As written, this reaction is …

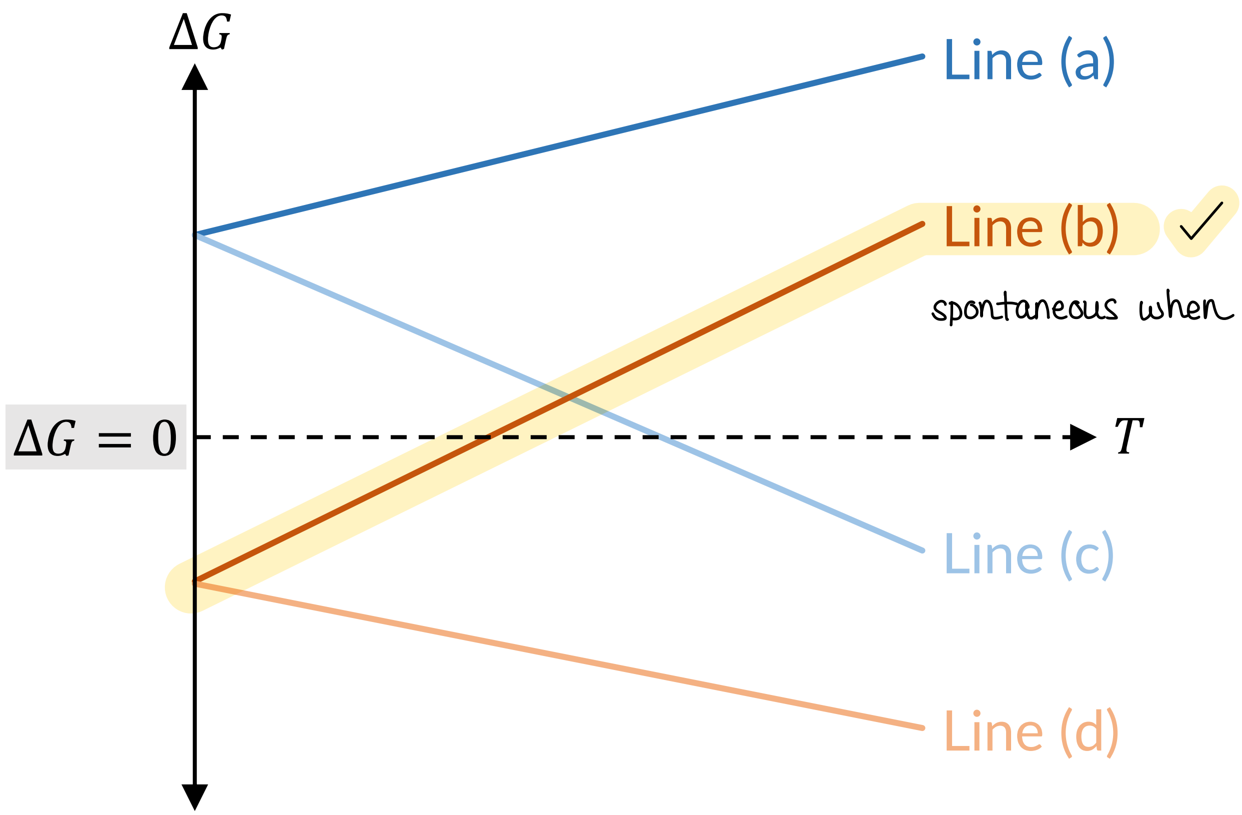
$$\underset{y}{\Delta G_{rxn}} = \underset{m}{(-\Delta S_{rxn})}\,\underset{x}{T} + \underset{b}{\Delta H_{rxn}}$$

- ☐ Spontaneous at all temperatures.
- ☑ Spontaneous at high temperatures.
- ☐ Spontaneous at low temperatures.
- ☐ Nonspontaneous at all temperatures.



IN-CLASS QUESTION 3

Learning Objective(s): 18.6, 18.7



spontaneous when T low!

Which line on the graph of ΔG versus T (given to left) correlates to the following hypothetical thermochemical reaction?

$$A(g) + B(g) \rightarrow AB(g) \qquad \Delta H_{rxn} < 0$$

$\Delta S_{rxn} < 0$, $\Delta n_{gas} < 0$

$$\underset{y}{\Delta G_{rxn}} = \underset{m}{(-\Delta S_{rxn})}\underset{x}{T} + \underset{b}{\Delta H_{rxn}}$$

Gibbs Free Energy and Equilibrium

18.6

Under standard conditions:

1 atm and 1 M

$Q = 1$

$$\Delta G^\circ = \Delta H^\circ_{rxn} - T\Delta S^\circ_{rxn}$$

very easy to calculate using ΔH°_f and S°

Under non-standard conditions:

$\neq$ 1 atm and $\neq$ 1 M

$Q \neq 1$

reaction quotient

$$\Delta G = \Delta G^\circ + RT \cdot \ln Q$$

When a reaction is at equilibrium:

$$0 = \Delta G^\circ_{rxn} + RT \cdot \ln K$$

$\Delta G = 0$ and $Q = K$

How to get K when equilibrium slow

$\Delta G^\circ_{rxn} = -RT \cdot \ln K$; $R = 8.314 \frac{J}{mol \cdot K}$

$$K = \exp\left(-\frac{\Delta G^\circ_{rxn}}{RT}\right)$$

- if $\Delta G^\circ_{rxn} < 1$, then $K > 1$
- if $\Delta G^\circ_{rxn} > 1$, then $K < 1$

$$\Delta H^\circ_{rxn} = \sum n_P \cdot \Delta H^\circ_{f,P} - \sum n_R \cdot \Delta H^\circ_{f,R} \qquad \Delta G^\circ_{rxn} = \sum n_P \cdot \Delta G^\circ_{f,P} - \sum n_R \cdot \Delta G^\circ_{f,R} \qquad \Delta G^\circ_{rxn} = -RT \cdot \ln K$$

EXAMPLE PROBLEM

$$\Delta S^\circ_{rxn} = \sum n_P \cdot S^\circ_P - \sum n_R \cdot S^\circ_R \qquad \Delta G^\circ_{rxn} = \Delta H^\circ_{rxn} - T\Delta S^\circ_{rxn} \qquad K = \exp\left(-\frac{\Delta G^\circ_{rxn}}{RT}\right)$$

Learning Objective(s): 18.7, 18.8, 18.9

$$R = 8.314 \frac{J}{mol \cdot K}$$

Calculate ΔG^o_{rxn} (in kJ/mol) and K_c for the following reaction at 25 °C.

$$Na^+(aq) + C_2H_3O_2^-(aq) \rightleftharpoons NaC_2H_3O_2(s)$$

ΔG^o_{rxn} = $23.5 \frac{kJ}{mol}$ $\qquad K_c$ = 7.62×10^{-5}

ΔH^o_{rxn} = $16.1 \frac{kJ}{mol}$ $\qquad \Delta S^o_{rxn}$ = $-24.9 \frac{J}{mol \cdot K}$

$$\Delta H^\circ_{rxn} = \left[(1)\left(-709.32 \frac{kJ}{mol}\right)\right] - \left[(1)\left(-240.3 \frac{kJ}{mol}\right) + (1)\left(-485.10 \frac{kJ}{mol}\right)\right] = 16.1 \frac{kJ}{mol}$$

$$\Delta S^\circ_{rxn} = \left[(1)\left(138.1 \frac{J}{mol \cdot K}\right)\right] - \left[(1)\left(58.5 \frac{J}{mol \cdot K}\right) + (1)\left(104.5 \frac{J}{mol \cdot K}\right)\right] = -24.9 \frac{J}{mol \cdot K}$$

$$\Delta G^\circ_{rxn} = \left[1 \cdot \Delta G^\circ_{f, NaC_2H_3O_2(s)}\right] - \left[1 \cdot \Delta G^\circ_{f, Na^+(aq)} + 1 \cdot \Delta G^\circ_{f, C_2H_3O_2^-(aq)}\right]$$

$$= \left[(1)\left(-607.7 \frac{kJ}{mol}\right)\right] - \left[(1)\left(-261.905 \frac{kJ}{mol}\right) + (1)\left(-369.3 \frac{kJ}{mol}\right)\right]$$

$$\Delta G^\circ_{rxn} = 23.5 \frac{kJ}{mol}$$

— or —

$$\Delta G^\circ_{rxn} = \Delta H^\circ_{rxn} - T\Delta S^\circ_{rxn} = 16.1 \frac{kJ}{mol} - (298.15 K)\left(0.0249 \frac{kJ}{mol \cdot K}\right) = 23.5 \frac{kJ}{mol}$$

$$K = \exp\left(-\frac{\Delta G^\circ_{rxn}}{RT}\right) = \exp\left[-\frac{23.5 \times 10^3 \frac{J}{mol}}{\left(8.314 \frac{J}{mol \cdot K}\right)(298.15 K)}\right] = 7.62 \times 10^{-5}$$

Substance	$\Delta H^o_f \left(\frac{kJ}{mol}\right)$	$S^o \left(\frac{J}{mol \cdot K}\right)$	$\Delta G^o_f \left(\frac{kJ}{mol}\right)$
$Na^+(aq)$	−240.3	+58.5	−261.905
$C_2H_3O_2^-(aq)$	−485.10	+104.5	−369.3
$NaC_2H_3O_2(s)$	−709.32	+138.1	−607.7

BONUS QUESTION

Assessed on: Summer 2023, Final Exam

The chemical reaction below is spontaneous at 800 K and nonspontaneous at 400 K.

$$AB_2(g) \rightleftharpoons A(g) + 2\ B(g)$$

Which statement is true?

- ❑ The reaction is endothermic.
- ❑ The reaction is exothermic.
- ❑ There is not enough information to determine if this reaction is endothermic or exothermic.